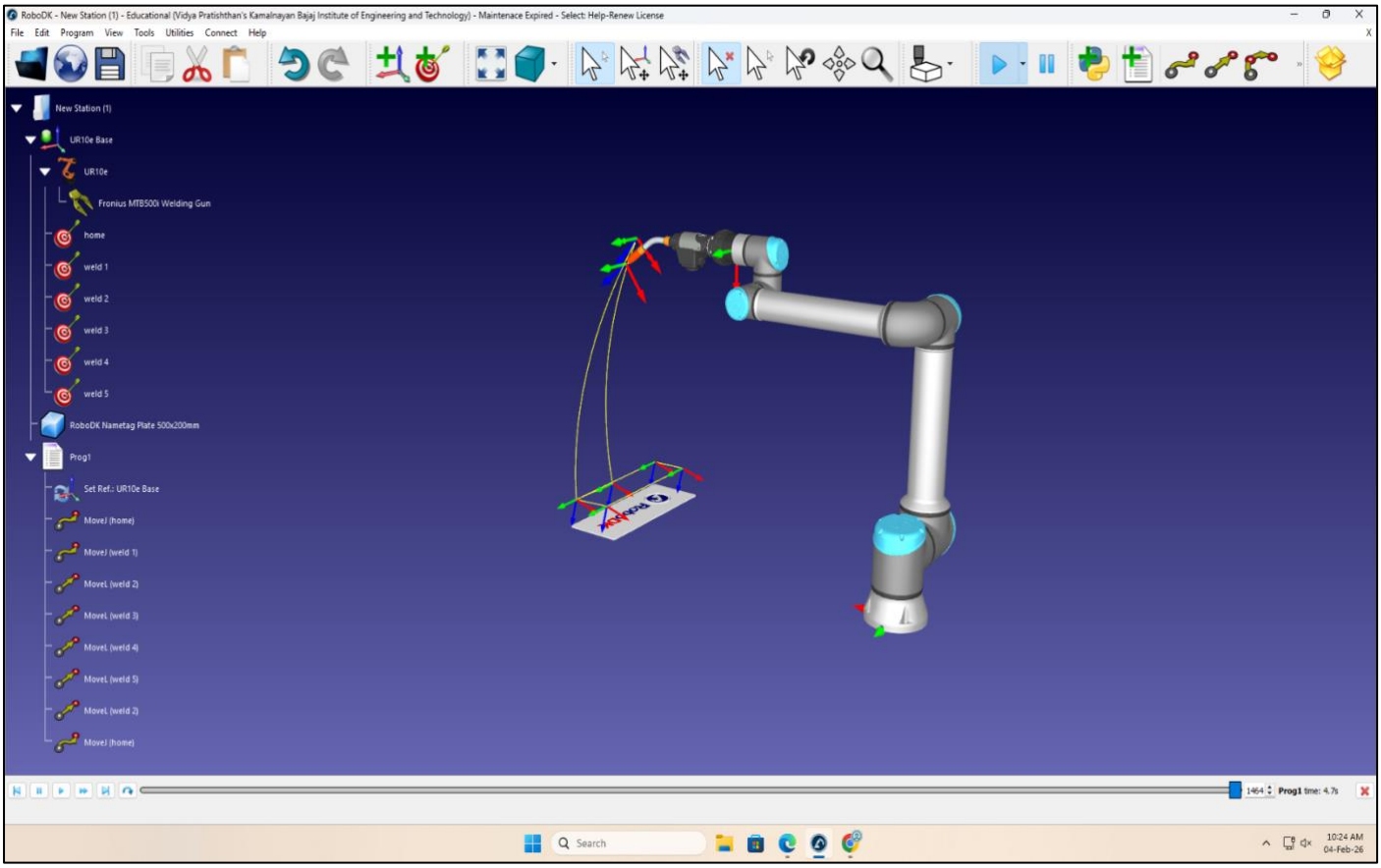
Experiment no :3

Title: Simulation of Spherical robot configuration using RoboDK Software.



- ❖ **Selected Robot:**

UR10e – Universal Robots

Why Selected?

- 6 DOF articulated robot
- High flexibility
- Suitable for welding applications
- Easy offline programming in RoboDK
- Industrial collaborative robot

- ❖ **Selection of Gripper:**

Tool Used:

Fronius MTB500i Welding Gun

Why?

- Specifically designed for robotic welding
- Accurate arc welding
- Suitable for metal nameplate welding

Tool is attached to robot flange.

- **Selection of objects:**

Object Used:

RoboDK Nameplate 500×200 mm

Purpose:

Robot performs welding along marked paths.

created multiple weld targets:

- weld 1
- weld 2
- weld 3
- weld 4
- weld 5

- **Explain the procedure of marking Targets:**

Steps Followed:

1. Insert reference frame on workpiece
2. Move robot to first weld point
3. Click Add Target
4. Rename target (weld 1, weld 2…)
5. Repeat for all weld points
6. Set orientation so torch angle is correct

- **Explain the procedure of marking Instructions:**

Procedure of Marking Instructions:

In Program

- MoveJ (home)

- MoveL (weld 1)
- MoveL (weld 2)
- MoveL (weld 3)
- MoveL (weld 4)
- MoveL (weld 5)
- MoveJ (home)

Why MoveL for Welding?

Because welding requires:

- Straight line motion
- Constant speed
- Smooth path

Instruction Types Used:

- MoveJ → for positioning
- MoveL → for welding path

❖ **Explain the procedure of marking programs:**

Steps:

- Right click robot → Add Program
- Add MoveJ to Home
- Add MoveL for each weld target
- Add final MoveJ to Home
- Run simulation

Simulation Result:

Robot follows curved yellow trajectory to perform weld path.